

SESSION 2

# Predictive Transportation Analytics

*ETA, Delay Risk & Exception Prediction*

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**73.62%**

Mail/Express punctuality  
(FY 2023-24, down from 90.48%  
in FY 2021-22)<sup>1</sup>

**27 of 33**

Punctuality-loss factors  
are internal to IR  
(CAG 2021)<sup>2</sup>

**₹32,378 Cr**

Annual traction energy  
bill in FY 2024-25 —  
idling = real money<sup>3</sup>

<sup>1</sup> PAC Report 2026 <sup>2</sup> CAG Report No. 22 of 2021 <sup>3</sup> Ministry of Railways FY24-25 data

# Session Roadmap — 3 Hours

## AGENDA

### SECTION 1

0–75 min

## Railway Logistics & Predictive Intuition

- › Business of punctuality
- › Network effect & cascading
- › Static timetable → dynamic ETA
- › How AI 'thinks': Features → Target

### SECTION 2

75–150 min

## Data-Driven Domain Understanding

- › Kaggle IR Train Delays Dataset
- › Visualising bottlenecks by zone
- › Correlation vs. causation
- › Decision Tree forecasting (light)

### SECTION 3

150–180 min

## Exception Management & Decision-Making

- Cost vs. service trade-off
- "The Corridor Crisis" scenario
- SLAs, passenger rights, law
- Mixed-team mitigation proposals

*Audience: IRMS Probationary Officers (mixed Arts/Legal/Science/Engineering backgrounds) • Focus: Business intuition over coding depth*

## Key Takeaways from This Session

1

Understand the business and operational costs of network delays in Indian Railways.

2

Grasp the basic intuition of predictive analytics without getting bogged down in complex coding.

3

Use open-source railway data to identify business bottlenecks and systemic inefficiencies.

4

Structure cross-functional business decisions (cost vs. service level) when AI predicts severe disruptions.

## SECTION 1

# Railway Logistics & Predictive Intuition

The business of punctuality, the network effect, and how AI translates operations into features → targets

75 MINUTES

# Indian Railways at a Glance — Scale of the Problem

FY 2023-24

**24,922**

Trains per day  
(13,198 pax + 11,724 freight)<sup>1</sup>

**720 Cr**

Passengers carried  
in FY 2023-24<sup>2</sup>

**1,588 MT**

Revenue freight loaded  
(₹1.6 L Cr revenue)<sup>1</sup>

**69,181 km**

Route length —  
99.2% BG electrified<sup>3</sup>

## Why this scale changes the analytics game

- › Every 1 percentage-point drop in punctuality = ~131 additional trains arriving late per day, each carrying revenue, crew, and follow-on rake commitments.
- › Indian Railways has more daily trains than the EU-27 + UK combined — the network effect amplifies every primary delay.
- › ₹60,466 Cr passenger subsidy means even small efficiency gains free up real fiscal headroom for safety + capex.

## (a) The Business of Punctuality — A Decade of Drift

PAC 2026

### MAIL/EXPRESS PUNCTUALITY

90.48%

*FY 2021-22 (post-COVID peak)<sup>1</sup>*

73.62%

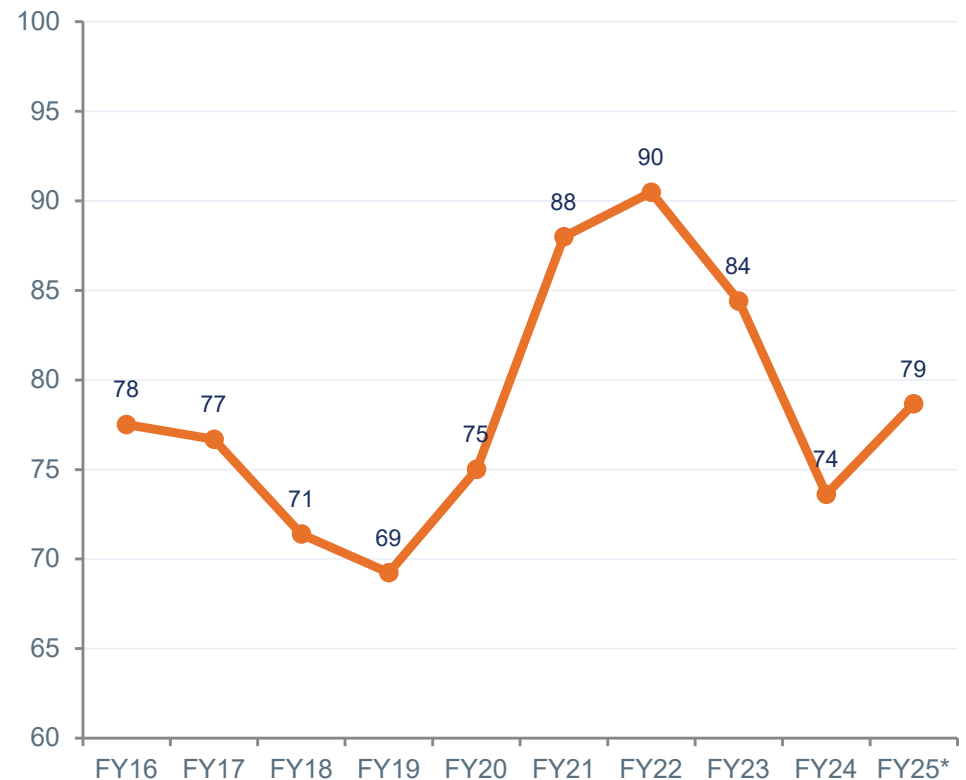
*FY 2023-24 — 17pp drop in 2 years<sup>1</sup>*

78.67%

*FY 2024-25 (up to Aug) — partial recovery<sup>1</sup>*

*PAC: punctuality is measured ONLY at the terminating station —  
Japan measures in seconds at every stop.*

### M/E Punctuality — Decadal Trend (%)



[1] Parliamentary Public Accounts Committee, Report on Punctuality (Lok Sabha, 4 Feb 2026); Business Standard 5 Feb 2026

# The Good News — Most Causes Are Within Our Control

CAG 2021

## 27 of 33

Punctuality-loss factors  
are internal to IR<sup>1</sup>

## 12.89%

External factors (weather,  
law & order) — only contributed  
this much over 2017-19<sup>1</sup>

## 65%

Of loss from just 6 causes:  
Out-of-path, Engineering,  
Rescheduling, Inter-zonal delays,  
Planned blocks, Traffic<sup>1</sup>

## 3.9× rise

Punctuality complaints to  
Rail Madad: 9,100 (FY16)  
→ 35,793 (FY18)<sup>1</sup>

Of loss from just 6 causes:

Out-of-path, Engineering,  
Rescheduling, Inter-zonal delays,  
Planned blocks, Traffic<sup>1</sup>

Punctuality complaints to  
Rail Madad: 9,100 (FY16)  
→ 35,793 (FY18)<sup>1</sup>

## What this means for predictive analytics

*If 87% of delay variance is internal, then 87% is predictable from operational data we already collect — controller actions, block schedules, upstream train movements. AI doesn't need clairvoyance; it needs a clean view of our own system.*

[1] CAG of India, Report No. 22 of 2021 — Performance Audit on Punctuality in Train Operations

# Premium ≠ Immune — The Vande Bharat Paradox

VANDE BHARAT — CY 2024

**+21%**

YoY rise in average delay<sup>1</sup>

**17 min**

avg delay  
(median 8 min)<sup>1</sup>

**105%**

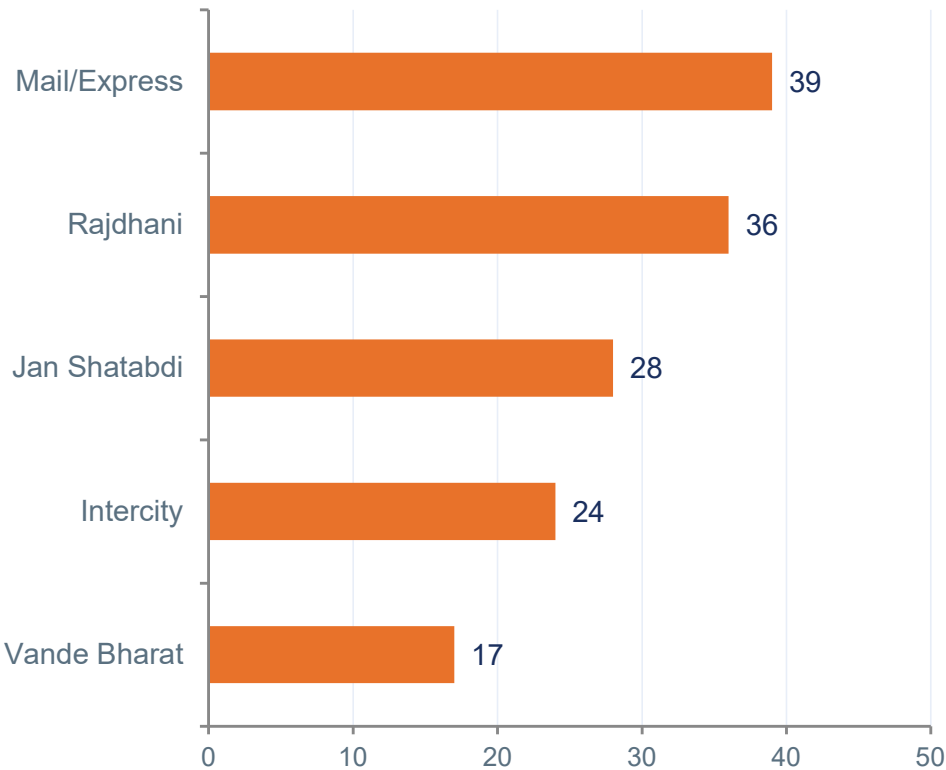
occupancy  
FY25-26 (to Jun)<sup>2</sup>

**164**

services running  
as of Sep 2025<sup>2</sup>

*Visibility raises stakes: every late premium service is a press headline.*

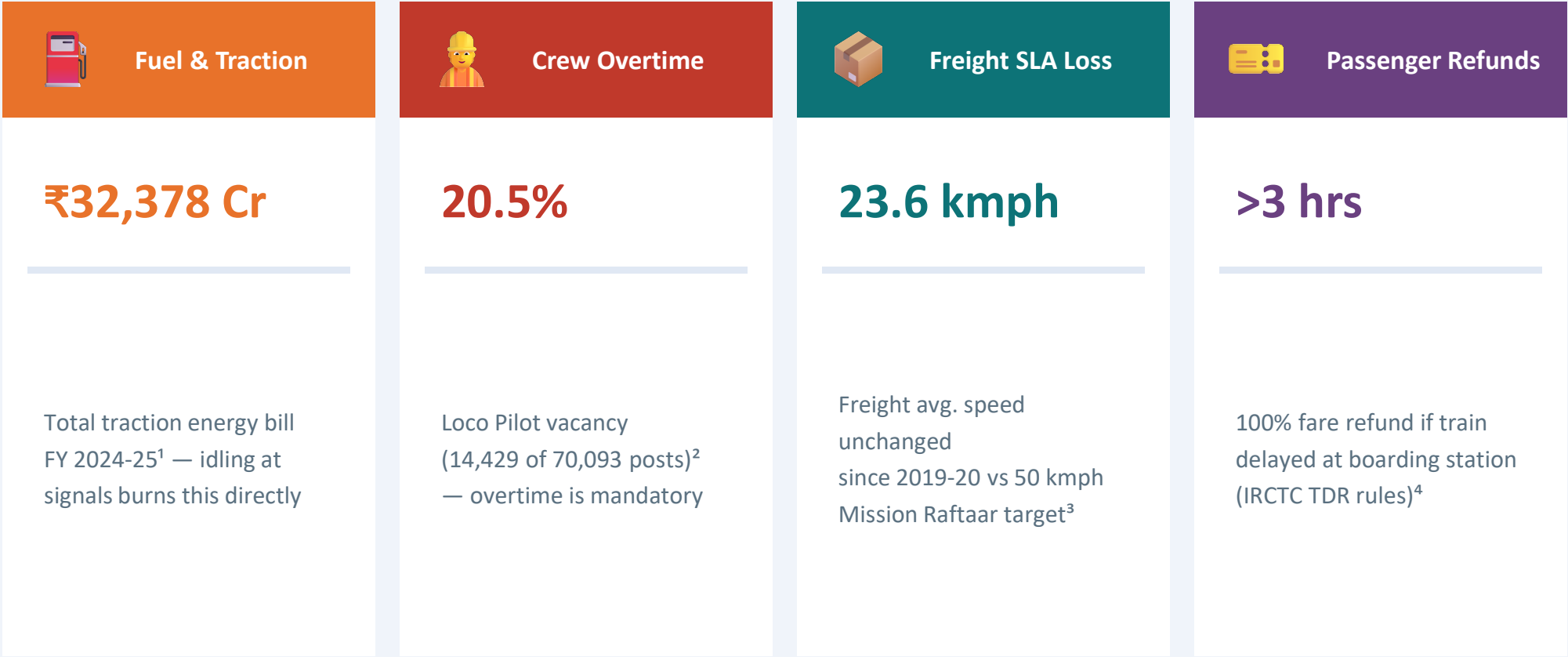
Average Delay by Train Type — CY 2024 (min)



[1] RailYatri Annual Delay Report 2024, via Business Standard (Jan 2025) [2] PIB Year-End Review 2024 (PRID 2088668); PIB Note 156964 (2025)

# Economic Cost of Delays — Where Money Leaks

Every minute of delay touches four cost buckets that already sit on IR's P&L:



[1] Ministry of Railways FY24-25 traction data via Angel One [2] Min. of Railways Lok Sabha reply (1 Mar 2024) on LP vacancies [3] Ministry of Railways Rajya Sabha reply Dec 2023; DFCCIL Corporate Plan [4] IRCTC Cancellation & Refund Rules, Para 7

# How the Rest of the World Monetises Delays

U K

## Schedule 8 Delay-Minute Regime

Every minute of delay attributed to Network Rail or TOC is monetised via Marginal Revenue Effect. ~14 M delay-minutes/year cost >£1 bn historically.

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*Delay Repay: 25%/50%/100%  
at 15/30/60 min thresholds<sup>1</sup>*

E U - 2 7

## Regulation 2021/782 Mandatory Compensation

Recast Rail Passenger Rights, in force 7 Jun 2023. Article 19: 25% of ticket price for 60-119 min; 50% for ≥120 min. Reimbursement within 30 days.

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*Force-majeure exemption  
for extreme weather<sup>2</sup>*

## IRCTC Refund Rules + Discontinued Pilot

100% fare refund if train delayed >3 h at boarding station and passenger doesn't travel. Tejas pilot (₹100/₹250 at 1h/2h) ran Oct 2019 – Feb 2024.

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*No standing compensation  
regime today<sup>3</sup>*

[1] ORR Schedule 8 Factsheet (Jun 2021); Network Rail Delay Attribution Board [2] Regulation (EU) 2021/782, Article 19 [3] IRCTC Rules; The Week 16 Apr 2026 on Tejas pilot discontinuation

## (b) The Network Effect — Why 10 Minutes Becomes 2 Hours

### How Delays Cascade — Stylised

#### Train A

Primary delay +10 min  
(loop crossing wait)



#### Train B (behind A)

Knock-on +25 min  
(headway compression)



#### Train C (cross-block)

Pathing conflict +45 min  
(single-line section)



#### Trains D, E, F

Cascading +120 min total  
across corridor

### Indian Corridor Cascade — In Numbers

# 4.19 Cr

*MINUTES of train detention on NR + NCR + ECR alone over 2017-19<sup>1</sup>*

#### Top 2 cascade drivers (CAG 2021):

- › Out-of-path running (path unavailability when train is ready)
- › Engineering & Planned blocks (track maintenance windows)

*Foundational theory: Goverde (2010) max-plus algebra; Carey & Kwieciński (1994) — reduced headways amplify knock-on delays non-linearly<sup>2,3</sup>*

## (c) From Static Timetables to Dynamic ETAs

| Dimension  | STATIC TIMETABLE<br>(Planning Reality)                                     | DYNAMIC ETA<br>(Operational Reality)                                              |
|------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Built from | Section speeds, dwell norms, path conflicts — set once per timetable cycle | Live train positions, upstream delays, weather, crew status — refreshed every 30s |
| Says       | "Train 12952 should reach Vadodara at 10:42"                               | "Train 12952 will reach Vadodara at 11:17, 92% confidence"                        |
| Updates    | Once a year (timetable revision)                                           | Continuously, with RTIS pings                                                     |
| Used by    | Passengers checking schedule, marketing                                    | Section controllers, freight customers, rake-link planners                        |
| Fails when | Reality diverges from plan (every day)                                     | Sensors fail or upstream data is stale                                            |
| IR's tools | Train at a Glance (TAG), Railways Time Table                               | NTES, RTIS-fed COA, RailDrishti <sup>1</sup>                                      |

[1] PIB PRID 1606967, 1861729; CRIS RTIS project page — 8,700+ locomotives RTIS-equipped

# How AI 'Thinks' — Features → Target

## FEATURES — What the model reads



### Day of week, holiday flag

*Mondays + festivals = higher delay*



### Scheduled departure hour

*Late-night vs peak morning differ*



### Monsoon / weather alert

*Konkan, NE under monsoon SOP*



### Upstream delay (primary)

*Already late at last station?*



### Train class & priority

*Rajdhani > Mail/Express > Passenger*



### Section congestion

*Trains/hour on this block now*



## TARGET — What the model predicts

### On Time

Within 15 min of scheduled  
(IR's official benchmark)<sup>1</sup>

### Slightly Delayed

15 min – 60 min late  
(operational nuisance)

### Severely Delayed

> 60 min late  
(exception management trigger)

*Marković et al. (2015) — Day-of-week, departure time, distance, upstream delay are the workhorse features<sup>2</sup>*

[1] Indian Railways punctuality definition — 15 minutes at terminating station [2] Marković et al. (2015) Transp. Res. Part C 56: 251-262 — Serbian Railways, 727 trains

## SECTION 1 — CHECKPOINT

# 5-Minute Reflection Before We Open Data

**Q1**

PAC says IR measures punctuality only at the terminating station. If you redesigned the metric tomorrow, what THREE additional measurement points would you add — and why?

**Q2**

Of the 4 cost buckets (fuel, crew, freight SLA, refunds), which one is most visible in YOUR division? Which is most invisible — and why is invisibility itself a risk?

**Q3**

Given that 87% of delay causes are internal (CAG), name one upstream operational signal in your zone that a predictive model could feed on TODAY.

*Think-Pair-Share (3 min) → Class debrief (2 min) → Coffee break before Section 2*

## SECTION 2

# Data-Driven Domain Understanding

Kaggle case study, descriptive → predictive analytics, and a light Decision Tree forecasting exercise

75 MINUTES

# Example: Train Delay Hotspot



## Hotspot Analysis

Where does delay actually concentrate?

## Indian Railways Train Delay Hotspots

*A 3-axis diagnostic — Zone × Time × Season*

Anchored to CAG 2021 + PAC 2026 + RailYatri 2024

[https://github.com/meettilak/IIMK-Tutorials/blob/main/session2/Train\\_Delay\\_Correlation\\_Hotspot\\_Analysis.ipynb](https://github.com/meettilak/IIMK-Tutorials/blob/main/session2/Train_Delay_Correlation_Hotspot_Analysis.ipynb)

# (a) The Kaggle 'Indian Railways Train Delays' Dataset

OPEN DATA

## PRIMARY DATASET FOR THIS SESSION

### Indian Railways Train Delays Dataset 2025

[kaggle.com/datasets/naijilaji/indian-railways-passenger-train-delays-dataset](https://kaggle.com/datasets/naijilaji/indian-railways-passenger-train-delays-dataset)

Passenger train arrival & departure data with delay attribution • Public, CSV format • Suitable for descriptive dashboards and light classification

#### Backup datasets (if the primary is unavailable on session day):

#### Kaggle Competition

Binary classification target: train  $\geq$  15 min late (matches IR definition)

[kaggle.com/competitions/indian-railways-predict-train-delay](https://kaggle.com/competitions/indian-railways-predict-train-delay)

#### DA323 IIT Guwahati

Curated Guwahati $\leftrightarrow$ Delhi/Mumbai/Chennai/Kolkata, Mar 2023–Mar 2024

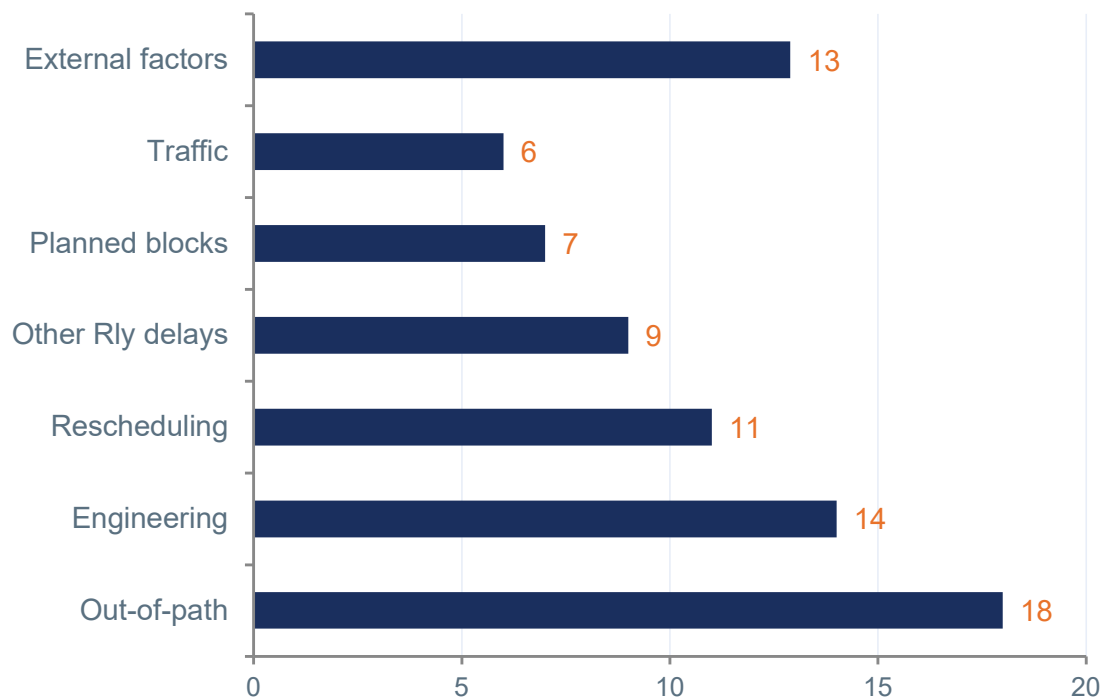
[github.com/ankitaanand28/DA323...](https://github.com/ankitaanand28/DA323_IndianRailwayTrainDelayDatasets)

[1] Kaggle dataset: [naijilaji/indian-railways-passenger-train-delays-dataset](https://kaggle.com/datasets/naijilaji/indian-railways-passenger-train-delays-dataset) (2025) [2] [ankitaanand28/DA323\\_IndianRailwayTrainDelayDatasets](https://github.com/ankitaanand28/DA323_IndianRailwayTrainDelayDatasets) — CC-BY-NC-SA

## (b) Visualising Bottlenecks — Where Are the Trouble Zones?

*From CAG 2021: top 6 internal causes account for 65% of all punctuality loss. Use the dataset to see which corridors carry the load.*

**Punctuality Loss by Cause (CAG 2021, 2017-19 sample)**



### Dashboard Questions to Ask the Data

1

Which zone has the worst median arrival delay?

2

Which corridor has the longest tail (P95 delay)?

3

Does delay scale linearly with distance — or only beyond a threshold?

4

Are monsoon months systematically worse, or only specific routes?

# Correlation vs. Causation — Two Stories the Data Tells

## CORRELATION

### Distance vs. Delay

Long-distance trains accumulate more absolute delay minutes — but per-100-km, late-night trains accumulate FASTER.

*Distance correlates with delay, but TIME-OF-DAY drives the per-km rate.*

## CAUSATION

### Upstream Delay → Downstream Delay

If Train A is 30 min late entering a single-line section, Train B that follows is 18-25 min late at the same section with ~85% probability.

*This is the cascade effect. It is mechanism, not coincidence (Goverde 2010).*

## SPURIOUS

### Vande Bharat = Delay Magnet?

Vande Bharat avg delay rose 21% YoY — but VB also runs on the most congested corridors (Delhi-Howrah, Mumbai-Ahmedabad).

*It's the CORRIDOR, not the train type. Always check confounders.*

## (c) Why Decision Trees First? — The Executive's Model

### Decision Trees in 30 Seconds

*A series of YES/NO questions, learned from data, that route every example to a prediction.*

#### Interpretable

Any officer can read the tree top-down without statistical training<sup>1</sup>

#### Auditable

Every prediction has a traceable rule chain — RTI-compliant

#### Fast

Trains in seconds on a laptop; runs in milliseconds in production

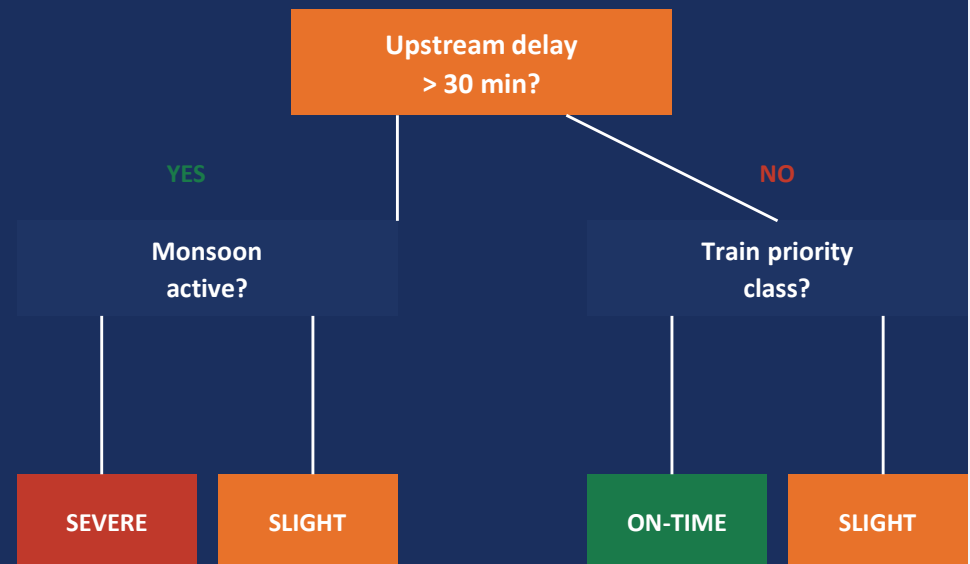
#### Handles mixed data

Numeric (delay, distance) + categorical (zone, day) — no preprocessing

#### Surfaces feature importance

Tells you which variables actually matter — a business signal in itself

### A Tree the Model Might Learn



*Anyone can read this tree in 30 seconds. That's the point.*

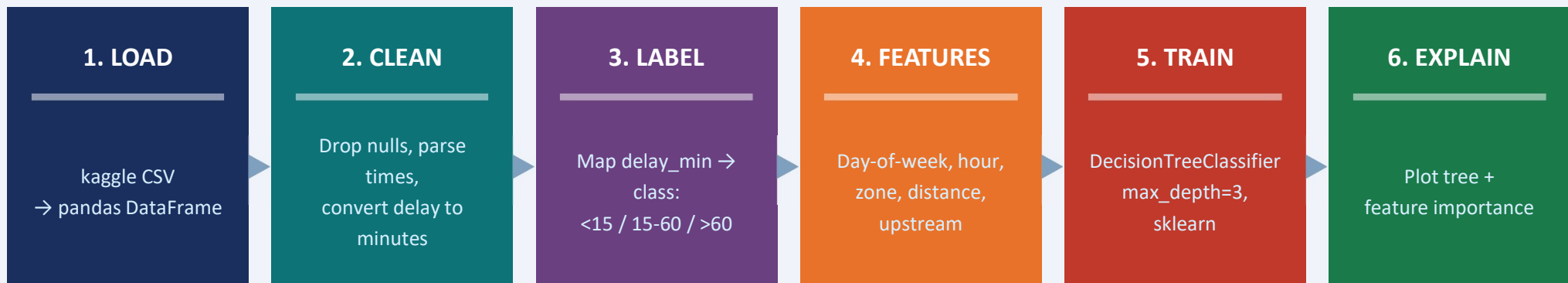
*Foundational: Breiman, Friedman, Olshen, Stone (1984) — CART<sup>2</sup>*

[1] Loh, W-Y. (2014). 'Fifty Years of Classification and Regression Trees,' Int. Statistical Review 82(3): 329-348 [2] Breiman, Friedman, Olshen, Stone (1984). Classification and Regression Trees. Wadsworth

# Light Hands-On — The Forecasting Setup

EXERCISE

*Goal: build a 3-class Decision Tree predicting whether a train will be On Time, Slightly Delayed, or Severely Delayed. Maximum depth = 3 (so the whole tree fits on one slide).*



```
# Minimal – all in one cell, no GPU, no deep-learning
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, plot_tree
```

```
df = pd.read_csv('train_delays.csv').dropna()
df['class'] = pd.cut(df['delay_min'], bins=[-1, 15, 60, 9999],
                    labels=['On Time', 'Slight', 'Severe'])
```

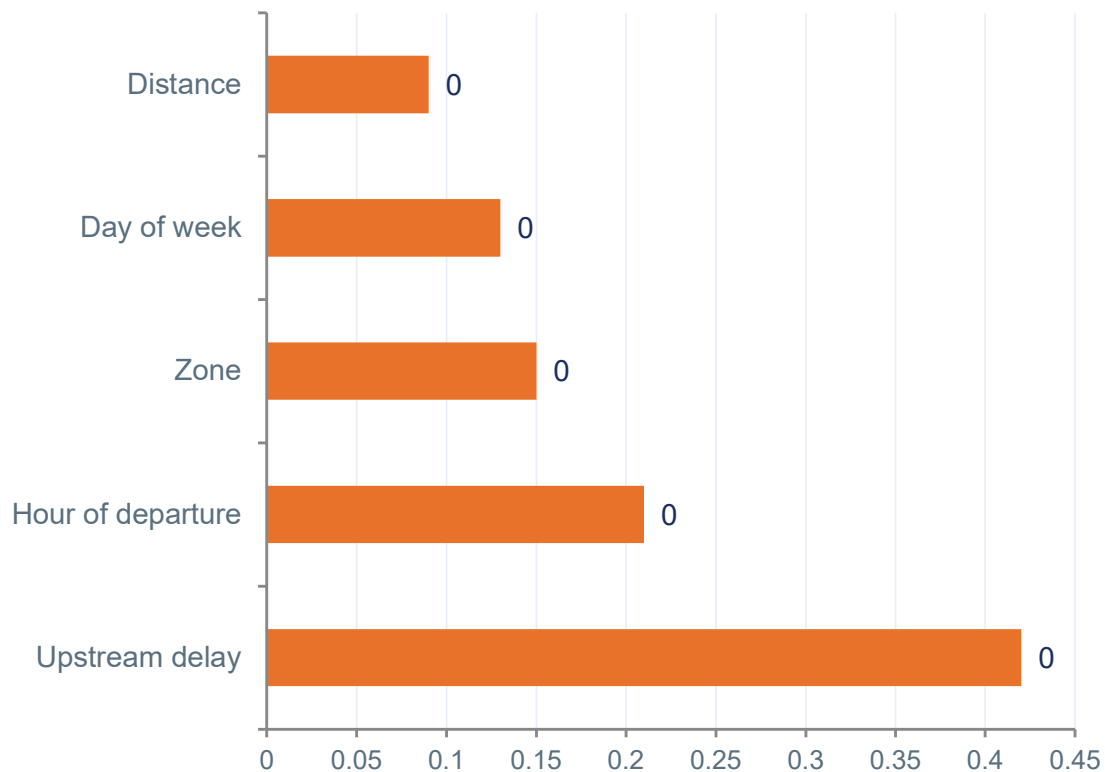
```
X = pd.get_dummies(df[['hour', 'dow', 'zone', 'distance_km', 'upstream_delay']])
```

[1] scikit-learn DecisionTreeClassifier — CART-based implementation (Pedregosa et al. 2011)

# Reading the Output — A Business Manager's View

## EXERCISE

Feature Importance — What the Tree Weighs



## Business Discussion

Q  
1

Upstream delay dominates (42%). Does this surprise you? It should not — it confirms the cascade.

Q  
2

Hour-of-departure matters more than distance. What operational lever does this suggest?

Q  
3

If 'Zone' is high, the model is partly learning institutional differences. Is that fair?

Q  
4

Where does the model say 'I don't need this feature'? That's where you can stop collecting it.

# From Notebook to Control Room — What IR Has Today

## RTIS

8,700+ locos

Real-Time Train Information System — GPS pings every 30s, integrated with NTES and Control Office Application (COA)<sup>1</sup>

## NTES

Public-facing

National Train Enquiry System — passenger-side ETA. Currently rule-based; ML overlay being piloted<sup>1</sup>

## RailDrishti

CRIS  
dashboard

Earnings, tickets, freight loading, punctuality, asset failures, bridge inspections — descriptive analytics layer<sup>2</sup>

## Kavach

1,452 RKM

Commissioned (Mar 2026) on Delhi-Mumbai and Delhi-Howrah. Future ETA enabler via continuous train-position telemetry<sup>3</sup>

### Commercial proof-point:

*RailYatri's self-learning ETA model delivers better accuracy on ~75% of trains compared to legacy rules-based ETAs (Kapil Raizada, RailYatri co-founder)<sup>4</sup>. The model uses unsupervised clustering on time-location data plus crowd-sourced GPS — exactly the kind of approach a Decision Tree generalisation can scale.*

[1] PIB PRID 1606967, 1861729; CRIS RTIS project page [2] CRIS e-Drishti / RailDrishti project pages [3] PIB PRID 2245720 (Mar 2026); DD News (Mar 2026) [4] Kapil Raizada (RailYatri) — LinkedIn article on ML ETA

## SECTION 2 — WRAP

# What We Just Built — And Where It Falls Short



A 3-class Decision Tree that any officer can read top-down — no statistics degree required.



Feature importance that confirms operational intuition: upstream delay dominates, monsoon and hour matter, distance is secondary.



An auditable, RTI-compliant pipeline — every prediction has a traceable rule chain.



Single tree = high variance. A real production system uses Random Forest or Gradient Boosting (see Pital et al. 2021 — RF best for IR data).



Past data  $\neq$  future. Concept drift (e.g., new VB launches, DFC roll-out) requires retraining every quarter.

# Data Rich modelling philosophy

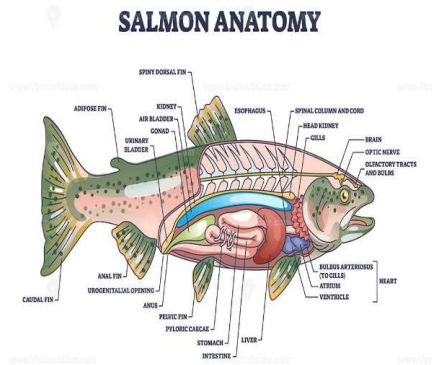
Check quality and prepare

Cleaning & Cutting

Flavour enrichment

Cook with ❤️

Taste it 😊 and... pay 💰



1

Prepare data for modelling and run data quality checks

2

Clean history for each train and remove black swan events

3

Enhance the training data with feature engineering

4

Train a model and use the model to generate forecasts

5

Forecast Evaluate KPIs, diagnosis & IT Cost management

# Future Fit AI Platform Reference Model

## Governance

**7. AI Policy - ethics, controls, performance standards and risk catalogues**

### **4. Data governance**

- Security
- Canonical master data model
- Quality metrics and usage stats
- Defined ownership of data objects
- Standard pathway and guardrails in the design and development of data in logical zones
- Data objects development capacity

**1. Business continuity and corporate risk strategy**

## Platform

### **9. AI Experience layer**

Self service, Persona tailored interfaces, APIs for third party connectivity and process integration, Semantic search, Chatbots, Agents

### **8. AI Devops and ML Ops**

### **6. AI Feature , Vector Stores, Graph Models**

Standardised feature generation for commonly used features  
Vector data bases to enable semantic search

### **5. Data lake with logical zones**

Raw / Bronze – immutable ingested data

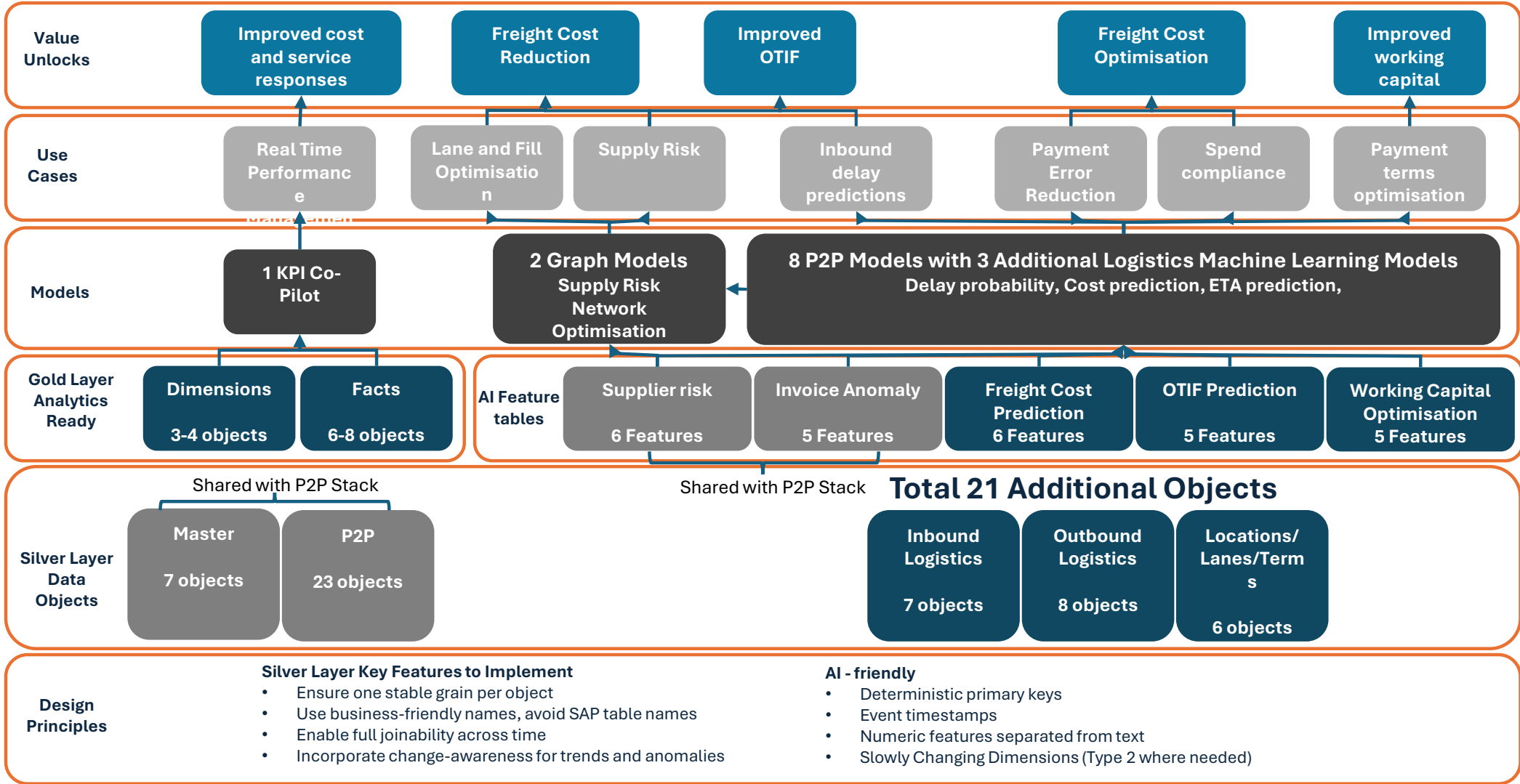
Refined / Silver – cleaned, conformed to canonical data model

Curated / Gold – AI-ready datasets with time travel capability

**3. Ingestion as batch and real time streaming including documents, images, sound and numerical data**

**2. Hybrid Cloud/on prem infrastructure for data platform and AI layers**

# Light House Example – Logistics Data & AI Stack



SECTION 3

# Exception Management & Business Decision-Making

Bridging the gap between the AI's prediction and the officer's operational response

30 MINUTES

## (a) The Cost vs. Service Trade-Off

When AI predicts a severe delay risk, the officer doesn't just 'react' — they CHOOSE between competing levers. Each lever has a cost.



### Reroute

*Send the train via an alternate path*

**Cost:** Higher fuel; longer journey

**Benefit:** Avoids congested section entirely



### Re-prioritise

*Prioritise this train over freight in the queue*

**Cost:** Freight SLA breach; loss of freight revenue

**Benefit:** Passenger service preserved



### Crew swap

*Pre-position fresh crew at next major station*

**Cost:** Crew overtime; HOER risk

**Benefit:** Avoid en-route detention waiting for crew



### Speed-up section

*Lift caution orders / speed restrictions if safe*

**Cost:** Marginal safety margin reduction

**Benefit:** Recover 5-15 min over 100 km



### Short-terminate

*End the train early; bus passengers onward*

**Cost:** Refunds + bus contracts + reputation hit

**Benefit:** Frees rake + crew for next service



### Accept the delay

*Let the delay propagate; manage communication*

**Cost:** Punctuality KPI hit; refund obligations

**Benefit:** No operational risk added

## ( b ) GROUP DECISION SCENARIO

# "The Corridor Crisis"

### THE SCENARIO

06:00 hrs Monday. Predictive model flags a 90% probability of a 4-hour cascading delay over the next 14 hours on a critical mixed-use corridor (5 Mail/Express + 2 Rajdhani + 4 freight rakes scheduled). Trigger: upstream Engineering block extended by 90 minutes due to a track-circuit failure at a junction station. Single-line section. Monsoon advisory active in the region. Two affected freight rakes carry premium coal (NTPC SLA).

### YOUR TEAM (mix of Arts, Legal, Science, Engineering) — 20 min

1. Propose a mitigation strategy using 2-3 of the 6 levers from the previous slide.
2. Defend it by balancing THREE forces: Operational Cost • SLAs • Public Policy / Legal
3. Specify what AI ALONE cannot decide — i.e., what's left for the human officer.

*Present 3 minutes per team. Faculty + peers vote on most defensible reasoning, not most optimal outcome.*

# The Three Forces Your Decision Must Balance

## OPERATIONAL COST

- Fuel & traction (₹32,378 Cr annual)<sup>1</sup>
- Crew overtime under HOER<sup>2</sup>
- Track usage charges
- Rake turnaround impact

## SERVICE LEVEL AGREEMENTS

- Premium freight SLAs (NTPC coal, SAIL steel)
- Vande Bharat punctuality KPI
- Inter-zonal performance contracts
- DFCCIL haulage commitments

## PUBLIC POLICY / LEGAL

- IRCTC 3-hour refund rule<sup>3</sup>
- Railway Claims Tribunal Act 1987
- Passenger safety regulations
- Consumer Protection Act applicability

# Reference Case — Konkan Railway Monsoon Playbook

KRCL 2024

## KRCL DISRUPTIONS 2024

### July 10:

Pernem tunnel waterlogged — 18-hour closure<sup>1</sup>

### July 14-15:

Landslide between Diwankhavati & Vinhere — multiple trains cancelled/diverted; state bus deployment<sup>1</sup>

*Two disruptions in ONE WEEK — predictive monsoon analytics is not optional, it is policy.*

## KRCL Monsoon SOP — Already Predictive<sup>2</sup>

672

Personnel patrolling the route during monsoon

9

Self-recording rain gauges (Mangaon to Udupi)

3

Flood warning systems (Kali, Savitri, Vashishti rivers)

40 kmph

Speed restriction in heavy rain — codified rule

10 Jun – 31 Oct

Monsoon timetable window every year

[1] India TV News, 15 Jul 2024; Konkan Katta News reports [2] Konkan Railway Corporation Ltd., 'KR Gears Up for Monsoon-2024' (konkanrailway.com/en/node/5076)

## KEY TAKEAWAYS

1

### **Delays are a P&L problem:**

Fuel + crew + freight SLAs + refunds = real money on IR's balance sheet. ₹32,378 Cr traction alone in FY24-25.

2

### **87% of causes are internal:**

CAG 2021: 27 of 33 punctuality-loss factors are within IR's control. That makes the problem PREDICTABLE — not predetermined.

3

### **Features beat algorithms:**

Upstream delay, hour of departure, monsoon flag, zone — these are 80% of the signal. A simple Decision Tree captures it.

4

### **Interpretability wins in govt.:**

RTI-compliant, auditable, single-screen models earn officer trust. Black-boxes lose policy battles before they win prediction ones.

5

### **Exception management is judgment:**

The "Corridor Crisis" has no optimal answer — only a defensible one. Balance Operational Cost, SLAs, and Public Policy / Legal.

## References (1/2) — Government & Industry Sources

### REFERENCES

1. Parliamentary Public Accounts Committee (2026). Report on Punctuality and Travel Time in Train Operations in Indian Railways. Lok Sabha, 4 Feb 2026.
2. CAG of India (2021). Performance Audit of Punctuality and Travel Time in Train Operations. Report No. 22 of 2021, Chapter 2. [cag.gov.in](https://cag.gov.in)
3. Ministry of Railways. Indian Railways Year Book 2023-24. Statistics & Economics Directorate, Railway Board. [rdso.indianrailways.gov.in](https://rdso.indianrailways.gov.in)
4. PIB Ministry of Railways (Mar 2025; Aug 2025). 'Most Railway Divisions Maintain Over 90% Punctuality' (PRID 2112484); 'Affordable Transportation to Over 720 Cr Passengers' (PRID 2154319).
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# Questions & Discussion

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## Session 2 Recap

- ✓ Business of punctuality — PAC, CAG, costs
- ✓ Network effect: 10-min → 2-hour cascade
- ✓ Static timetables → dynamic ETAs
- ✓ Kaggle dataset, Decision Tree exercise
- ✓ "Corridor Crisis" scenario
- ✓ 28+ cited primary & academic sources

## Next: Session 3

Introduction, Applications overview,  
Data & KPIs to consider (Dr. Alok Patel)